Fair Test Practical · Tier Pack · Aspire

Learning goal: I can record repeated results, spot an odd reading, and use the data to answer our question.

Aspire Route (Book First)

Explain why excluding the flagged anomaly (1.3 s) gives a more accurate mean for the 30x30 cm canopy, and evaluate stopwatch reaction-time error.

Help: Including the tangled 1.3 s drop would pull the mean down unfairly. Stopwatch reaction time (~0.2 s) matters most on the __ fall times.

Optional Make It · Aspire

Make an Uncertainty & Improvement Audit Card

Calculate the range of each set of repeats (max minus min) and propose two method improvements to reduce timing error.

Word Help

air resistance: a frictional force from air pushing against a moving object

repeat trials: doing the exact same test more than once to check the readings match

anomaly: an odd result that is very different from the other repeats

conclusion: a scientific answer that uses the recorded numbers to explain the pattern

Exit Check

Which parachute took the longest time to fall: 10x10 cm (1.2 s) or 30x30 cm (3.4 s)?

Why does a larger parachute canopy fall more slowly?

What should you do if one repeat trial is an anomaly because the strings tangled?